



Operating System : Scheduling Algorithms

Scheduling

Algorithm

- Basic Concepts
- First-Come, First-Served (FCFS) Scheduling Algorithm
- Shortest-Job-First (SJF) Scheduling Algorithm
 - Nonpreemptive SJF Scheduling
 - Preemptive SJF Scheduling
- Priority Scheduling Algorithm
 - Nonpreemptive Scheduling Algorithm
 - Preemptive Scheduling Algorithm
- Round-Robin Scheduling Algorithm
- Advantages of Round-Robin Scheduling Algorithm
- Disadvantages of Round-Robin Scheduling Algorithm

Scheduling Algorithm

- The scheduling algorithm deals with the problem of deciding which of the process in the ready queue is to be allocated the CPU.
- There are many CPU scheduling algorithm as shown below.
 - First-Come, First- Served (FCFS) scheduling algorithm
 - Shortest -Job –First (SJF) scheduling algorithm
 - Priority scheduling algorithm
 - Round- Robin scheduling algorithm

FCFS Scheduling Algorithm

- First-Come, First-Served Scheduling Algorithm is nonpreemptive algorithm.
- It is the simplest of all the scheduling algorithms.
- The key concept of this algorithm is “The process which comes first in the ready queue will allocate the CPU first”.
- The next process will allocate the CPU only after the previous gets fully executed.

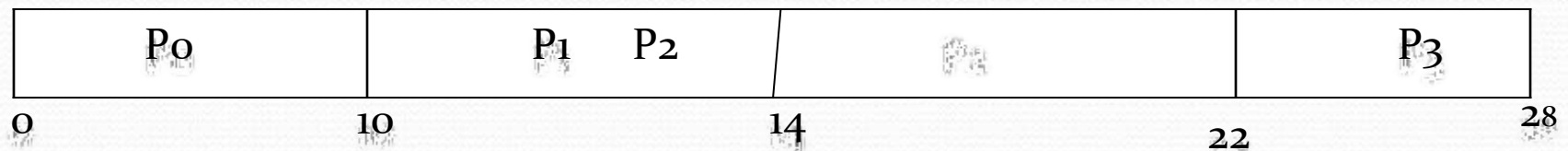
FCFS Scheduling Algorithm

Example- 1:

Sr. No	Process	Execution Time/ Burst Time(ms)
1	p0	10
2	p1	4
3	p2	8
4	p3	6

If the arrival time is not given. We can assume arrival time of all process is 0.

Gantt Chart:



FCFS Scheduling Algorithm

Completion Time:

$P0 = 10, P1 = 14, P2 = 22, P3 = 28$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

$P0 : 10 - 0 = 10 \text{ ms}$

$P1 : 14 - 0 = 14 \text{ ms}$

$P2 : 22 - 0 = 22 \text{ ms}$

$P3 : 28 - 0 = 28 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(10+14+22+28) / 4 = 18.5 \text{ ms}$

FCFS Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

$$P0 : 10 - 10 = 0 \text{ ms}$$

$$P1 : 14 - 4 = 10 \text{ ms}$$

$$P2 : 22 - 8 = 14 \text{ ms}$$

$$P3 : 28 - 6 = 22 \text{ ms}$$

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

$$\text{Average Waiting Time(AWT)} = (0+10+14+22) / 4 = 11.5 \text{ ms}$$

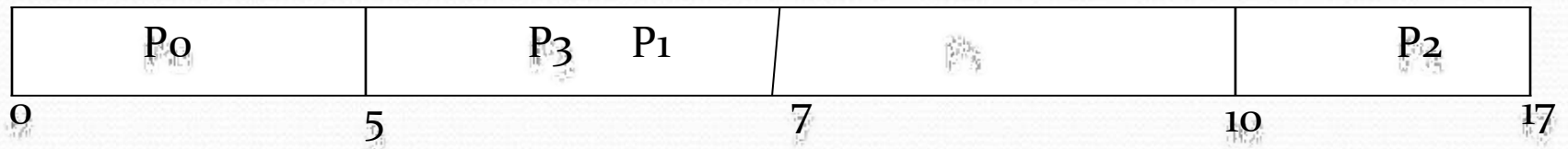
FCFS Scheduling Algorithm

Example- 2:

Sr. No	Process	Arrival Time	Execution Time/ Burst Time(ms)
1	p0	0	5
2	p1	4	3
3	p2	6	7
4	p3	2	2

When Arrival Time is given

Gantt Chart:



FCFS Scheduling Algorithm

Completion Time:

$P0 = 5, P1 = 10, P2 = 17, P3 = 7$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

$P0 : 5 - 0 = 5 \text{ ms}$

$P1 : 10 - 4 = 6 \text{ ms}$

$P2 : 17 - 6 = 11 \text{ ms}$

$P3 : 7 - 2 = 5 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(5+6+11+5) / 4 = 6.75 \text{ ms}$

FCFS Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

$$P0 : 5 - 5 = 0 \text{ ms}$$

$$P1 : 6 - 3 = 3 \text{ ms}$$

$$P2 : 11 - 7 = 4 \text{ ms}$$

$$P3 : 5 - 2 = 3 \text{ ms}$$

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

$$\text{Average Waiting Time(AWT)} = (0+3+4+3) / 4 = 2.5 \text{ ms}$$

SJF Scheduling Algorithm

- Shortest Job First scheduling(SJF) works on the process with the shortest **burst time**.
- SJF is the best approach to minimize waiting time.
- SJF is a Greedy Algorithm.
- In case of a tie, it is broken by FCFS scheduling algorithm.
- Predicting the time the process will use on its next schedule:

$$t(n+1) = w * t(n) + (1 - w) * T(n)$$

Here: $t(n+1)$ is time of next burst.

$t(n)$ is time of current burst.

$T(n)$ is average of all previous bursts .

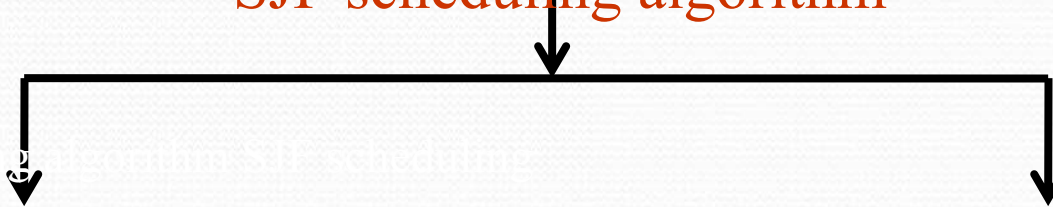
w is a weighting factor emphasizing current or previous bursts.

Types of SJF Scheduling

Algorithm

- SJF scheduling algorithm can be categorized into two parts.

SJF scheduling algorithm



Non-Preemptive SJF

Preemptive SJF

Shortest Remaining Time First
(SRTF)

SJF Scheduling

Algorithm

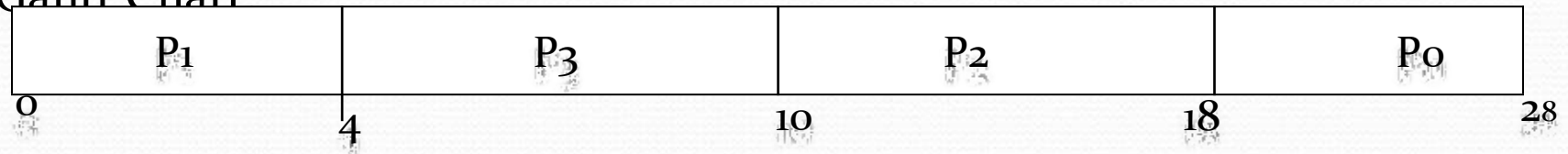
- The SJF algorithm may be either preemptive or non-preemptive
- A preemptive SJF will preempt this currently executing process and starts the execution of newly entered process.
- Nonpreemptive SJF will allow the currently executing process to complete its burst time without any interruption in its execution.
- Preemptive SJF scheduling is sometimes called Shortest-Remaining-Time-First (SRTF) scheduling

Nonpreemptive SJF Scheduling Algorithm

Sr. No	Process	Execution Time/ Burst Time(ms)
1	p0	10
2	p1	4
3	p2	8
4	p3	6

If the arrival time is not given. We can assume arrival time of all process is 0.

Gantt Chart:



Nonpreemptive SJF Scheduling Algorithm

Completion Time:

$P_0 = 28, P_1 = 4, P_2 = 18, P_3 = 10$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

$P_0 : 28 - 0 = 28 \text{ ms}$

$P_1 : 4 - 0 = 4 \text{ ms}$

$P_2 : 18 - 0 = 18 \text{ ms}$

$P_3 : 10 - 0 = 10 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(28+4+18+10) / 4 = 15 \text{ ms}$

Nonpreemptive SJF Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

P0 : $28 - 10 = 18$ ms

P1 : $4 - 4 = 0$ ms

P2 : $18 - 8 = 10$ ms

P3 : $10 - 6 = 4$ ms

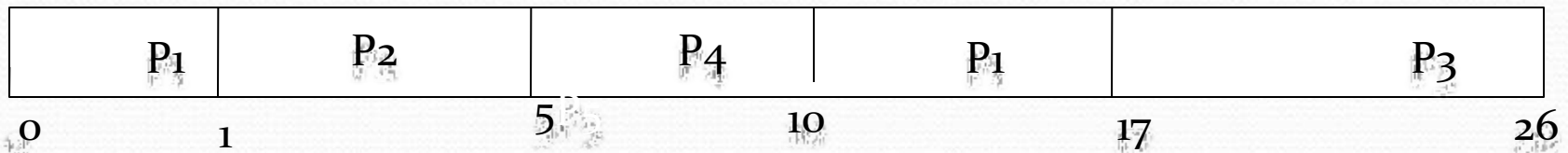
Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

Average Waiting Time(AWT) = $(18 + 0 + 10 + 4) / 4 = 8$ ms

Preemptive SJF Algorithm

Sr. No	Process	Arrival Time	Execution Time/ Burst Time(ms)
1	p0	0	8
2	p1	1	4
3	p2	2	9
4	p3	3	5

Gantt Chart



Preemptive SJF Scheduling Algorithm

Completion Time:

$P_0 = 17, P_1 = 5, P_2 = 26, P_3 = 10$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

$P_0 : 17 - 0 = 17 \text{ ms}$

$P_1 : 5 - 1 = 4 \text{ ms}$

$P_2 : 26 - 2 = 24 \text{ ms}$

$P_3 : 10 - 3 = 7 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(17+4+24+7) / 4 = 13 \text{ ms}$

Preemptive SJF Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

P0 : $17 - 8 = 9$ ms

P1 : $4 - 4 = 0$ ms

P2 : $24 - 9 = 15$ ms

P3 : $7 - 5 = 2$ ms

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

Average Waiting Time(AWT) = $(9 + 0 + 15 + 2) / 4 = 6.5$ ms

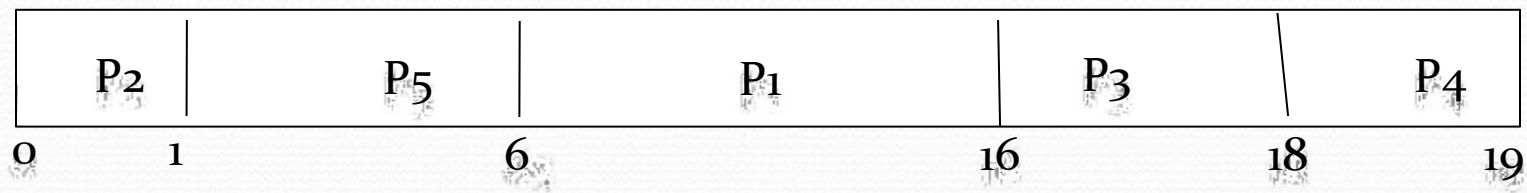
Priority Scheduling

- A priority is associated with each process and the CPU is allocated to the process with the highest priority.
- Equal priority processes are scheduled in FCFS order.
- Priorities are generally some fixed range of numbers such as 0 to 7. some system represents low numbers to represent low priority while others use low numbers to represent high priority.

Priority Scheduling Algorithm

Sr. No	Process	Burst time	Priority
1	P ₁	10	3
2	P ₂	1	1
3	P ₃	2	4
4	P ₄	1	5
5	P ₅	5	2

Gantt Chart:



Priority Scheduling Algorithm

Completion Time:

P1 = 16, P2 = 1, P3 = 18, P4 = 19, P5 = 6

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)

Turnaround Time(TAT) of each process

P1 : $16 - 0 = 16$ ms

P2 : $1 - 0 = 1$ ms

P3 : $18 - 0 = 18$ ms

P4 : $19 - 0 = 19$ ms

P5 : $6 - 0 = 6$ ms

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(16+1+18+19+6) / 5 = 12$ ms

Priority Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

$$P1 : 16 - 10 = 6 \text{ ms}$$

$$P2 : 1 - 1 = 0 \text{ ms}$$

$$P3 : 18 - 2 = 16 \text{ ms}$$

$$P4 : 19 - 1 = 18 \text{ ms}$$

$$P5 : 6 - 5 = 1 \text{ ms}$$

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

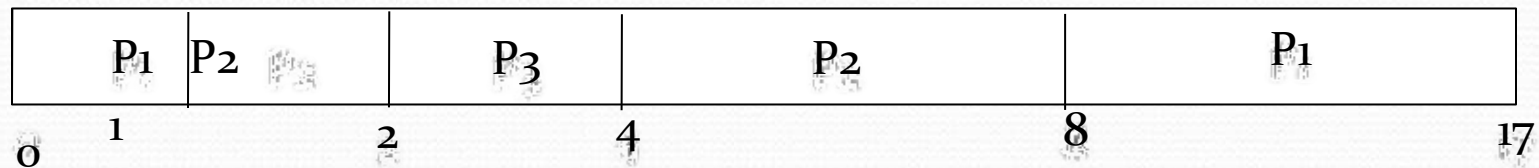
$$\text{Average Waiting Time(AWT)} = (6+0+16+18+1) / 5 = 8.2 \text{ ms}$$

Note:- Priority Scheduling is Either Preemptive or Non-preemptive

Preemptive Priority Scheduling

Sr. No	Process	Burst time	Priority	Arrival Time
1	P ₁	10	3	0
2	P ₂	5	2	1
3	P ₃	2	1	2

Gantt Chart:



Priority Scheduling Algorithm

Completion Time:

$P1 = 17, P2 = 8, P3 = 4$

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT) Turnaround Time(TAT) of each process

$P1 : 17 - 0 = 17 \text{ ms}$

$P2 : 8 - 1 = 7 \text{ ms}$

$P3 : 4 - 2 = 2 \text{ ms}$

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(17+7+2) / 3 = 8.6 \text{ ms}$

Priority Scheduling Algorithm

Waiting Time(WT) of Each process = Turn Around Time(TAT) – Burst Time(BT)

P1 : $17 - 10 = 7$ ms

P2 : $7 - 5 = 2$ ms

P3 : $2 - 2 = 0$ ms

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

Average Waiting Time(AWT) = $(7+2+0) / 3 = 3$ ms

Note:-

- if the two process having the same priority then process with shorter burst time will be executed.
- If the process having the same burst time then process will be executed on the basis of FCFS scheduling.

Starvation and

Ageing

- **Starvation** or indefinite blocking is phenomenon associated with the Priority scheduling algorithms, in which a process ready to run for CPU can wait indefinitely because of low priority.
- To avoid starvation, we use the concept of **Aging**. In Aging, after some fixed amount of time quantum, we increase the priority of the low priority processes. By doing so, as time passes, the lower priority process becomes a higher priority process.

Round-Robin Scheduling

Algorithm

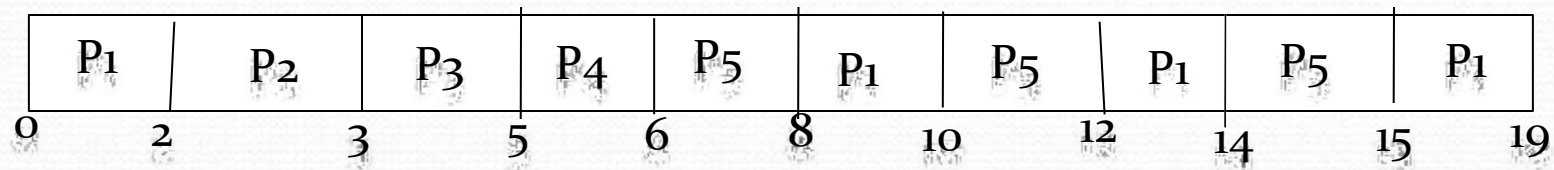
It is designed especially for time sharing systems. Here CPU switches between the processes. When the time quantum expired, the CPU switched to another job. A small unit of time, called a time quantum or time slice. A time quantum is generally from 10 to 100 ms. The time quantum is generally depending on OS. Here ready queue is a circular queue. CPU scheduler picks the first process from ready queue, sets timer to interrupt after one time quantum and dispatches the process.

Round-Robin Scheduling Algorithm

Sr. No	Process	Burst time
1	P ₁	10
2	P ₂	1
3	P ₃	2
4	P ₄	1
5	P ₅	5

Time Quantum is 2
ms.

Gantt Chart:



Round-Robin Scheduling

Algorithm

Completion Time:

P1 = 19, P2 = 3, P3 = 5, P4 = 6, P5 = 15

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT) Turnaround Time(TAT) of each process

P1 : $19 - 0 = 19$ ms

P2 : $3 - 0 = 3$ ms

P3 : $5 - 0 = 5$ ms

P4 : $6 - 0 = 6$ ms

P5 : $15 - 0 = 15$ ms

Average Turnaround Time(ATAT) = Total turn around time of all processes / Total no of processes

Average Turnaround Time(ATAT) = $(19+3+5+6+15) / 5 = 9.6$ ms

Round-Robin Scheduling

Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst Time(BT)

Waiting Time(WT) of each process

$$P1 : 19 - 10 = 9 \text{ ms}$$

$$P2 : 3 - 1 = 2 \text{ ms}$$

$$P3 : 5 - 2 = 3 \text{ ms}$$

$$P4 : 6 - 1 = 5 \text{ ms}$$

$$P5 : 15 - 5 = 10 \text{ ms}$$

Average Waiting Time(AWT) = Total waiting time of all processes / Total no of processes

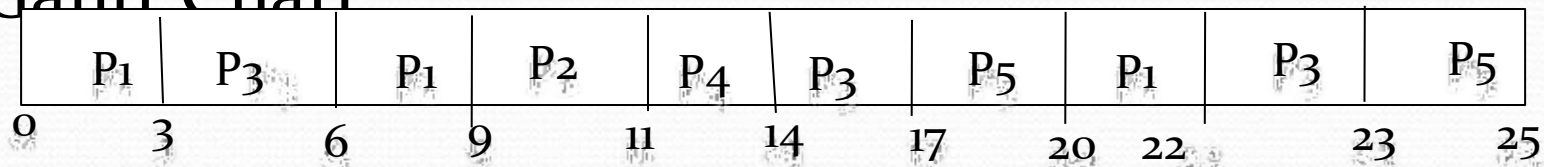
$$\text{Average Waiting Time(AWT)} = (9+2+3+5+10) / 5 = 5.8 \text{ ms}$$

Round-Robin Scheduling Algorithm

S.No.	Process	Arrival Time (AT)	Burst Time (BT)
1	P ₁	0	8
2	P ₂	5	2
3	P ₃	1	7
4	P ₄	6	3
4	P ₅	8	5

Time Quantum (TQ)= 3
ms.

Gantt Chart.



Round-Robin Scheduling

Algorithm

Completion Time:

P1 = 22, P2 = 11, P3 = 23, P4 = 14, P5 = 25

Turnaround Time(TAT) = Completion Time(CT) – Arrival Time(AT)
Turnaround Time(TAT) of each process

P1 : $22 - 0 = 22$ ms

P2 : $11 - 5 = 6$ ms

P3 : $23 - 1 = 22$ ms

P4 : $14 - 6 = 8$ ms

P5 : $25 - 8 = 17$ ms

Average Turnaround Time(ATAT) = Total turn around time of all processes /
Total no of processes

Average Turnaround Time(ATAT) = $(22+6+22+8+17) / 5 = 15$ ms

Round-Robin Scheduling Algorithm

Waiting Time(WT) = Turnaround Time(TAT) – Burst

Time(BT) Waiting Time(WT) of each process

$$P1 : 22 - 8 = 14 \text{ ms}$$

$$P2 : 6 - 2 = 4 \text{ ms}$$

$$P3 : 22 - 7 = 15 \text{ ms}$$

$$P4 : 8 - 3 = 5 \text{ ms}$$

$$P5 : 17 - 5 = 12 \text{ ms}$$

Average Waiting Time(AWT) = Total waiting time of all processes /

Total no of processes

$$\text{Average Waiting Time(AWT)} = (14+4+15+5+12) / 5 = 10 \text{ ms}$$

Advantages of Round-Robin Scheduling

- It doesn't face the issues of starvation effect.
- All the jobs get a fair allocation of CPU.
- It deals with all process without any priority
- This scheduling method does not depend upon burst time. That's why it is easily implementable on the system.
- Once a process is executed for a specific set of the period, the process is preempted, and another process executes for that given time period.
- It gives the best performance in terms of average response time.

Disadvantages of Round-Robin Scheduling

- If slicing time of OS is low, the processor output will be reduced.
- This method spends more time on context switching
- Its performance heavily depends on time quantum.
- Priorities cannot be set for the processes.
- Round-robin scheduling doesn't give special priority to more important tasks.
- Lower time quantum results in higher the context switching overhead in the system.